

Oyster Reef Restoration Optimization

Site Selection under a JARS Metapopulation Connectivity Model

Project Report

Marouf Paul

M.S. Computational Operations Research, William & Mary

Advisor: Dr. Rex Kincaid • Collaborators: Dr. Leah Shaw, Dr. Rom Lipcius (VIMS)

Repository: `NetworkAnalysisOysters`

1. Purpose of This Document

This report has two jobs:

- Describe the problem, the JARS metapopulation model, the optimization formulations, and the heuristic algorithms in enough detail that the code is reproducible without reading every script.
- Report only verified results. Every number in the tables below was re-derived from the repository by re-running the JARS ODE on the final selected sets, and every cross-model intersection was recomputed from the saved CSVs.

Throughout the report I use the following convention: greedy, post-greedy 1-swap, and stingy backward elimination are evaluated under the full JARS equilibrium objective $F(S)$, while all MIQP variants optimize a quadratic plumbing surrogate $\tilde{F}(S)$ using AMPL and Gurobi. Both views are needed because the surrogate is fast and provable but lossy, and the ODE is faithful but expensive. The surrogate retains self-recruitment (the connectivity diagonal), matching the JARS recruitment term, whose internal sum runs over all sites including the reef itself.

2. Problem Statement and Biology

2.1 Restoration as a network problem

The Chesapeake Bay historically supported more than ten billion oysters; today the population is below 1% of that baseline. Restoration is expensive, so the question is not whether to restore but *where*: among 49 candidate reef sites distributed across five ecological management zones in the bay, which subset of $K = 25$ sites should be restored to maximize long-run adult oyster biomass?

The non-trivial part is that oyster larvae disperse via tidal currents. A reef's value depends not only on its own productivity but on the larvae it receives from, and sends to, the other selected

reefs. This makes the problem combinatorial and non-additive: pairs and clusters of reefs interact super-linearly, and restoring an isolated high-productivity reef can be worse than restoring three weaker but well-connected ones.

2.2 The JARS metapopulation ODE

Each candidate reef is modeled by a coupled four-state ODE for juveniles J , adults A , reef shell volume R , and sediment S (the JARS model, originally implemented by Prof. Leah Shaw and converted to Python in `src/model/jars_ode.py`). The state of reef k evolves as larval supply is filtered through density-dependent settlement, juveniles mature into adults, adults accrete shell, and sediment slowly buries shell. The key coupling term is the larval input

$$\Lambda_k = P_0(k) + \sum_{\ell} P_1(\ell \rightarrow k) |A_{\ell}|^{1.72},$$

where $P_0(k)$ is the external larval supply at site k , P_1 is the larval transport probability matrix between sites (the internal sum includes $\ell = k$, i.e. self-recruitment), and the exponent 1.72 (a published parameter for oysters) makes dense, retentive clusters disproportionately valuable. This nonlinearity is the structural reason greedy heuristics undershoot and clustered selection methods do well.

2.3 The optimization objective

Let $A_k^*(S)$ be the equilibrium adult biomass at site k when the ODE is integrated to $t = 1000$ with reefs restored at the sites in S . The true objective is

$$\max F(S) = \sum_{k \in S} A_k^*(S), \quad \text{subject to } |S| = 25, S \subseteq N, |N| = 49.$$

Every greedy / swap / stingy result reported below was scored by literally integrating this ODE on the chosen set. The MIQP models optimize a quadratic surrogate of F that captures the dominant external + internal-retention contributions; the alignment between the two is excellent in practice but is not assumed in the heuristic results.

3. Connectivity Data and External Forcing

3.1 Two connectivity matrices

VIMS collaborators provided two independent connectivity matrices from different hydrodynamic realizations:

- **Matrix 1:** `nk_All_060102final_56sites_Model.xlsx`
- **Matrix 2:** `nk_All_060103final_56sites_Model.xlsx`

Both matrices include 56 raw site labels. Per biological advice from Prof. Shaw and Prof. Lipcius, sites 66–72 are excluded, leaving 49 candidate sites for both matrices (`CANDIDATE_SITES` in

`src/model/jars_ode.py`). For the MIQP, P_1 is additionally scaled by `P1scaling = 0.5` and fed through the surrogate weight $W(\ell, k) = P_1(\ell, k) (A^*)^{1.72}$ with $A^* = 0.05675$ representing the median single-reef adult equilibrium. The diagonal $W(k, k)$ is retained, so self-recruitment is credited in the surrogate exactly as it is in the ODE.

3.2 Indexing convention (important)

There are two coexisting indexing schemes in this project, and confusing them is the easiest way to get wrong results.

Biological site labels are the original integer IDs in the Excel matrices: 1, 3, 4, 5, $\dots$, 60 (with sites 66–72 dropped, and several others in the 1–60 range absent). Every CSV file and every results table in this report uses these labels.

AMPL indices are 0, 1, $\dots$, 48, i.e. positions in the `CANDIDATE_SITES` array. Every `.dat` file under `ampl/` uses these indices. The translation between the two lives in `runs/oyster_index_mapping_matrix{1,2}.csv`.

In particular, the community sets in `ampl/oyster_comm.dat` (e.g. $C_1 = \{33, 36, 44\}$) are AMPL *indices*, not site labels. Looking up those indices in the mapping CSV shows that C_1 actually contains site labels $\{42, 48, 56\}$. The Python runner scripts (`scripts/run_miqp*.py`) handle the translation back to site labels before writing CSVs, so end-user output is always in biological labels.

3.3 External larval supply: two regimes

The same set of 49 sites is studied under two different external-forcing assumptions:

- **Constant** P_0 — every site receives the same external input of 170 larvae per timestep. This is the “pure plumbing” regime: differences in performance come only from the internal connectivity network, so this regime tells us about the structure of the network itself.
- **Realistic** P_0 — a heterogeneous vector with values in $\{100, 200, 400\}$ assigned per region based on empirical estimates (`setP0` in `jars_ode.py`, supplied by the biologists). This regime mixes “faucet strength” into the optimization: some sites are inherently well-fed even if they are weakly connected.

All MIQP results in this report are under constant P_0 (the AMPL data file embeds 170 for all sites). The realistic regime is only used for the ODE-driven heuristics.

4. Optimization Methods

4.1 ODE-driven heuristics

Greedy forward selection

Start with the empty set. At each step, evaluate $F(S \cup \{k\})$ by running the full JARS ODE for every candidate k not yet selected, and add the site with the largest marginal gain. Repeat 25

times. This is a fast first pass but myopic: it cannot recover from a poor early choice, and the $|A|^{1.72}$ nonlinearity means it tends to undershoot the best clustered design.

Post-greedy 1-for-1 local swap

Starting from the greedy 25-site set, evaluate every swap that exchanges one selected site for one unselected candidate. Apply the best improving swap, then repeat until no swap improves F . Empirically this lifts the greedy score by 1–2% in absolute terms and is what we treat as the heuristic upper bound.

Stingy backward elimination

Start with all 49 candidates. At each step, evaluate $F(S \setminus \{a\})$ for every $a \in S$, and remove the site whose removal hurts F the least. Stop when $|S| = 25$. This is a separate path to the local optimum, and is useful as a consistency check on the swap result — the two methods converging on the same 25 sites is strong evidence that the set is a stable local optimum.

4.2 MIQP surrogates

All MIQP variants share the surrogate objective

$$\max \tilde{F}(S) = \sum_{i \in S} Pe(i) + \sum_{i, j \in S} W(i, j)$$

with binary selection variables x_i and the cardinality constraint $\sum x_i = 25$. The double sum includes the diagonal term $W(i, i)$, which credits self-recruitment. There are four variants.

(a) Base MIQP — `oyster_quad.mod`

Just the cardinality constraint. Picks the most retentive 25-site cluster.

(b) MIQP + community minimums — `oyster_comm.mod`

Adds five geographic-region constraints: $|S \cap C_r| \geq r_r$ with $(r_1, r_2, r_3, r_4, r_5) = (2, 5, 3, 2, 3)$. The total floor of 15 leaves 10 “flex” sites for the optimizer to place freely. This is the equity model used to satisfy multi-stakeholder coverage.

(c) MIQP + variable sizing — `oyster_size.mod`

Adds a continuous reef-area variable $s_i \in [L_i, U_i]$ for each selected site (5–50 acres), with a global budget $\sum s_i \leq T = 1000$ acres and a normalization $\bar{S} = 20$. The objective scales each $Pe(i)$ and $W(i, j)$ entry by $s_i/\bar{S}$. This lets the model allocate big reefs to hubs and small reefs to marginal sites.

(d) MIQP + community + sizing — `oyster_comm_size.mod`

Combines (b) and (c). This is the model that most closely matches real planning constraints — it must include some sites for regional equity, but it can size those sites small to limit the cost in surrogate larvae production.

5. Results

All ODE scores below were independently re-derived from the repository by running `src/opt/evaluator.py` on the saved site sets. All ten heuristic scores match the original logs exactly to six decimal places. MIQP objectives are read from the AMPL/Gurobi logs. The site-set intersections in §5.3–5.4 were also recomputed.

5.1 Heuristic results under constant P_0

Table 1 lists the 25-site sets and equilibrium scores produced by the three ODE-driven heuristics on each matrix under constant external supply.

Table 1: ODE-driven heuristic results under constant P_0 .

Matrix / Method	Selected sites (sorted)	$F(S)$
M1 Greedy	4, 10, 15, 16, 17, 19, 20, 21, 26, 27, 28, 30, 31, 32, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54	1.846640
M1 Greedy + Swap	10, 12, 15, 16, 17, 20, 21, 26, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54, 59	1.880054
M1 Stingy	10, 12, 15, 16, 17, 20, 21, 26, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54, 59	1.880054
M2 Greedy	4, 10, 11, 12, 15, 17, 19, 20, 21, 26, 27, 28, 29, 30, 31, 32, 36, 37, 40, 41, 47, 49, 51, 52, 53	1.692871
M2 Greedy + Swap	10, 11, 12, 15, 16, 17, 19, 20, 26, 29, 31, 32, 33, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54, 59	1.733033
M2 Stingy	10, 11, 12, 15, 16, 17, 19, 20, 26, 29, 31, 32, 33, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54, 59	1.733033

Two structural observations from Table 1:

- On both matrices the swap improves on greedy by exchanging sites: on Matrix 1 it trades four sites ($\{4, 19, 27, 30\}$ for $\{12, 29, 33, 59\}$), raising F from 1.846640 to 1.880054 (+1.8%); on Matrix 2 it trades five sites ($\{4, 21, 27, 28, 30\}$ for $\{16, 33, 44, 54, 59\}$), raising F from 1.692871 to 1.733033 (+2.4%). The gain comes from changing the selected set, not from reordering a fixed one.
- On both matrices, post-greedy local swap and stingy backward elimination converge to exactly the same 25 sites with exactly the same equilibrium score. This is strong (independent-path) evidence that these sets are stable local optima of the true ODE objective.

5.2 Heuristic results under realistic P_0

Switching to heterogeneous external supply pulls some new sites into the solution (those with high P_0 even if weakly connected) and pushes some plumbing-only sites out. Table 2 shows the greedy + swap and stingy results.

Table 2: ODE-driven heuristic results under realistic P_0 .

Matrix / Method	Selected sites (sorted)	$F(S)$
M1 Greedy + Swap (real)	4, 6, 10, 15, 19, 20, 21, 24, 27, 30, 31, 32, 36, 37, 38, 39, 40, 41, 47, 49, 51, 52, 53, 55, 60	1.861968
M1 Stingy (real)	4, 10, 15, 16, 17, 19, 20, 21, 26, 27, 31, 32, 33, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54, 59	1.819074
M2 Greedy + Swap (real)	1, 4, 6, 10, 18, 19, 20, 21, 24, 30, 31, 35, 36, 37, 38, 39, 40, 41, 42, 47, 49, 53, 55, 57, 60	1.793457
M2 Stingy (real)	4, 6, 7, 10, 15, 16, 19, 20, 21, 24, 26, 27, 30, 31, 32, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53	1.723331

Notice that under realistic forcing, greedy + swap and stingy no longer agree on a single set — the heterogeneity of P_0 creates more local optima. However, a large “robust core” of sites is shared across all four realistic designs (see §5.4).

5.3 MIQP surrogate results under constant P_0

All four AMPL/Gurobi models solved to proven global optimality (nonconvex bilinear objectives, `nonconvex=2`) in well under a second on a standard laptop. Table 3 summarizes the objective values and selected sets.

Table 3: MIQP surrogate results under constant P_0 (self-recruitment retained, global optimum).

Model	Selected sites (sorted)	$\tilde{F}(S)$
M1 Base MIQP	10, 11, 12, 15, 16, 17, 20, 26, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 49, 51, 52, 53, 54, 59	14785.03
M1 MIQP + Comm	10, 12, 15, 16, 17, 21, 26, 28, 29, 31, 32, 33, 37, 40, 41, 42, 44, 47, 48, 49, 51, 52, 53, 54, 59	13844.54
M1 MIQP + Size	10, 11, 12, 15, 16, 17, 20, 26, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 49, 51, 52, 53, 54, 59	58564.19
M1 MIQP + Comm + Size	10, 12, 15, 16, 17, 21, 26, 28, 29, 31, 32, 33, 37, 40, 41, 42, 44, 47, 49, 51, 52, 53, 54, 56, 59	56443.65
M2 Base MIQP	10, 11, 12, 15, 16, 17, 20, 26, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 49, 51, 52, 53, 54, 59	16446.59
M2 MIQP + Comm	10, 11, 12, 15, 16, 17, 21, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 42, 47, 49, 51, 52, 53, 56, 59	15218.16
M2 MIQP + Size	10, 11, 12, 15, 16, 17, 20, 26, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 49, 51, 52, 53, 54, 59	66552.23
M2 MIQP + Comm + Size	10, 11, 12, 15, 16, 21, 26, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 42, 47, 48, 49, 51, 52, 53, 59	64448.16

Two notes on the sizing models. First, in both matrices the optimal *set* under sizing is identical to the base MIQP set — the optimizer doesn’t change who is selected, it just allocates large area (40–50 acres) to high-connectivity hubs and small area (5–15 acres) to weaker neighbors. Second, the total reef area is exactly 1000 acres in every sizing solution; the budget constraint is tight in all

cases.

Representative size profile (M1 MIQP + Size): 50 acres at sites 10, 11, 12, 15, 16, 17, 26–33; 40 acres at sites 40, 41, 44, 51–54, 59; and 5–15 acres at sites 20, 36, 37, 49. See `runs/miqp_size_sites_matrix{1,2}.csv` for the full per-site allocations.

5.4 Backbone sites: intersections across models and matrices

The point of running so many models is to see which sites are robust to the choice of method, matrix, and forcing regime. The intersection structure is the cleanest summary.

Within-matrix constant- P_0 intersection (all 7 constant models)

Matrix	Sites common to greedy, swap, stingy, MIQP, +Comm, +Size, +Comm+Size	Size
Matrix 1	10, 15, 16, 17, 26, 28, 31, 32, 37, 40, 41, 44, 49, 51, 52, 53, 54	17
Matrix 2	10, 11, 12, 15, 29, 31, 32, 36, 37, 40, 41, 49, 51, 52, 53	15

Matrix 1 has a 17-site backbone and Matrix 2 a 15-site backbone across the seven constant-forcing optimization runs.

Cross-matrix constant- P_0 backbone

Selection	Sites	Size
Intersection of M1 (17) and M2 (15) backbones	10, 15, 31, 32, 37, 40, 41, 49, 51, 52, 53	11

Global backbone (all 18 models, both matrices, both P_0 regimes)

Selection	Sites	Size
Sites selected by every one of the 18 model runs	10, 31, 37, 40, 41, 49, 53	7

These seven sites — **10, 31, 37, 40, 41, 49, 53** — appear in every greedy, every greedy+swap, every stingy, and every MIQP variant (base, +Comm, +Size, +Comm+Size) on both connectivity matrices and under both constant and realistic external forcing. They are the structural arteries of the bay’s larval network: any restoration design that excludes them is structurally weaker. (The 18 designs are the six constant- P_0 heuristics, the four realistic- P_0 heuristics reported in Table 2, and the eight MIQP solves. Standalone greedy under realistic forcing is myopic and drops site 37; it is used only as the seed for the realistic swap and is not counted among the 18.)

5.5 Network centrality on the constant- P_0 backbone

To explain why the same sites keep being selected, we computed five weighted centrality metrics on the directed connectivity graph defined by P_1 for each matrix: in-strength, out-strength, betweenness

(with edge length 1/weight), eigenvector centrality, and PageRank. Sites were ranked separately by each metric, and the average rank gives an overall “network importance” ordering. Tables 4 and 5 show the top 10 sites of the 11-site cross-matrix backbone for each matrix.

Table 4: Matrix 1 — top 10 of the backbone by average rank.

Site	In-str.	Out-str.	Betw.	Eig.	PR	avg rank
10	359,474	214,957	0.0266	0.5974	0.0999	7.0
40	292,701	218,022	0.0541	0.6304	0.0936	4.8
32	198,442	226,987	0.0966	0.1119	0.0273	6.0
31	173,926	144,740	0.0545	0.2975	0.0584	7.8
41	166,988	193,584	0.0306	0.2877	0.0456	10.0
53	93,500	123,635	0.0842	0.0466	0.0270	11.2
52	93,545	153,534	0.0563	0.0078	0.0138	15.8
15	194,082	179,547	0.0098	0.0206	0.0175	16.4
49	79,623	95,582	0.0381	0.0174	0.0237	16.6
37	120,416	20,909	0.0168	0.1453	0.0481	17.0

Table 5: Matrix 2 — top 10 of the backbone by average rank.

Site	In-str.	Out-str.	Betw.	Eig.	PR	avg rank
31	195,232	179,626	0.0408	0.3092	0.0750	8.8
40	423,838	115,054	0.0363	0.6913	0.1574	9.2
10	541,264	90,109	0.0355	0.5889	0.1311	10.2
49	130,852	155,422	0.0918	0.0178	0.0200	10.4
36	119,622	163,586	0.0395	0.1133	0.0346	12.2
41	161,622	234,275	0.0013	0.2140	0.0477	12.8
37	120,277	37,213	0.0758	0.1031	0.0323	13.0
53	87,729	196,023	0.0472	0.0183	0.0144	13.2
32	224,566	162,872	0.0186	0.0290	0.0144	15.0
15	159,499	156,282	0.0337	0.0094	0.0115	17.2

The pattern is consistent across both matrices: sites 10, 31, 32, 40, 41 are simultaneously high on eigenvector centrality and PageRank, meaning they are embedded in dense subnetworks of other important nodes. Sites 36, 49, 53 score relatively high on betweenness, indicating they act as bridges between otherwise weakly-connected regions. This is the network-theoretic explanation for why optimization keeps selecting them: the JARS dynamics reward retentive loops and bridges, both of which these sites provide.

6. Repository Structure and Reproducibility

6.1 Layout


```

NetworkAnalysisOysters/
|-- ampl/ AMPL .mod and .dat files
| |-- oyster_quad.mod Base MIQP
| |-- oyster_comm.mod MIQP + community minimums
| |-- oyster_size.mod MIQP + variable sizing
| |-- oyster_comm_size.mod MIQP + community + sizing
| '-- oyster_*.dat Auto-generated data (prepare_miqp_data.py)
|-- data/ Connectivity matrices (Excel)
|-- src/
| |-- model/jars_ode.py JARS ODE, candidate site list, P0 setter
| |-- opt/evaluator.py Single-call ODE scoring of a site set
| |-- opt/greedy.py Forward greedy
| |-- opt/local_search.py 1-for-1 swap local search
| |-- opt/backward.py Stingy backward elimination
| '-- utils/io_utils.py CSV/TXT writers for runs/
|-- scripts/ Entry-point CLIs (run_*.py, prepare_*.py)
|-- runs/ Saved CSVs, summaries, network metrics
|-- figures/ Generated PNGs
|-- validate.py Independent re-validation harness
'-- requirements.txt

```

6.2 Environment

- Python 3.10. `pip install -r requirements.txt` covers numpy, pandas, scipy, openpyxl, matplotlib, networkx, amply.
- AMPL + Gurobi on PATH. Test with `python -m scripts.check_ampl_env`.

6.3 End-to-end reproduction

All published numbers can be regenerated with the following sequence (substitute `-matrix 2` to repeat for the second connectivity matrix):

```

# 1. Generate AMPL data from the Excel matrix (diagonal/self-recruitment kept)
python -m scripts.prepare_miqp_data --matrix 1

# 2. ODE-driven heuristics (constant P0)
python -m scripts.run_greedy
python -m scripts.run_greedy_then_local
python -m scripts.run_backward

# 3. MIQP variants (constant P0)
python -m scripts.run_miqp --matrix 1
python -m scripts.run_miqp_comm --matrix 1
python -m scripts.run_miqp_size --matrix 1
python -m scripts.run_miqp_comm_size --matrix 1

# 4. Network centrality
python -m scripts.network_core_analysis --matrix 1

# 5. Figures
python -m scripts.make_figures

```

Run times on a recent laptop: each MIQP variant solves in well under a second; each greedy or

stingy run takes $\sim 4\text{--}5$ minutes (because every step calls the ODE); a full greedy + local swap run takes $\sim 15\text{--}30$ minutes depending on how many improving passes are needed. A combined driver, `run_everything.py`, parallelizes the per-candidate ODE evaluations across CPU cores and runs all heuristics and all eight MIQP solves in one pass; the parallel evaluation is bit-identical to the serial library functions.

6.4 Validation script

`validate.py` at the repository root is the independent re-validation harness for the handover. It reloads every reported design, re-runs the ODE on it, and recomputes all backbone intersections from scratch. Running it produces `runs/validation_summary.csv` and `runs/validation_network_core_matrix{1,2}`. Every number in this report came from that script.

7. Limitations and Recommended Next Steps

7.1 Limitations of the current setup

- **Equilibrium-only objective.** $F(S)$ is the sum of A at $t = 1000$. Transient dynamics, time-to-recovery, and resilience to shocks are not considered.
- **Fixed JARS parameters.** The 14 published JARS parameters are fixed; no robustness analysis was done over uncertainty in those parameters.
- **Surrogate weight uses a fixed A^* .** $W(i, j) = P_1(i, j) (0.05675)^{1.72}$ uses a single median single-reef equilibrium for A^* . A state-dependent surrogate could tighten the alignment between MIQP and ODE further, at the cost of solvability.
- **Community partition is exogenous.** The five communities come from geography, not from the connectivity graph. Data-driven community detection (Louvain, Infomap) on P_1 might produce more natural equity buckets.

7.2 Self-recruitment sensitivity

The canonical surrogate keeps self-recruitment (the connectivity diagonal). As a sensitivity check, zeroing the diagonal leaves the base and size *selections* unchanged on both matrices and lowers the base objective by about 7% on Matrix 1 ($14785 \rightarrow 13692$) and 4% on Matrix 2 ($16447 \rightarrow 15807$). Only the community-constrained selections and the Matrix 1 within-matrix backbone shift; the cross-matrix and seven-site global backbones are unaffected. The headline result is therefore robust to whether self-recruitment is credited in the surrogate, while the canonical model keeps it for consistency with the ODE.

7.3 Recommended next experiments

- **Robust / stochastic MIQP.** Treat the connectivity matrix entries as uncertain and solve a chance-constrained or two-stage stochastic version. The handful of disagreements between Matrix 1 and Matrix 2 are exactly the sites where this would matter.

- **Multi-objective frontier.** Trace the Pareto frontier of (biomass) vs. (equity tightness) vs. (total acreage). The current four models are four points on that frontier; ten or twenty would tell stakeholders what they’re really trading.
- **Phased restoration.** Replace the one-shot $|S| = 25$ selection with a multi-period plan that respects an annual restoration capacity. This would change the optimal sequence even if the final set stays similar.
- **Coupled with non-larval services.** The current objective is adult biomass. A more honest objective would also credit water filtration, fish habitat, and economic value at each site, weighted by stakeholder preferences.

7.4 Small cleanups for whoever takes over

- `scripts/network_core_analysis.py` hard-codes a core-site list at the top. The validated constant- P_0 backbones from this re-analysis are: global 7 sites (10, 31, 37, 40, 41, 49, 53), cross-matrix 11 sites, and within-matrix 17 (M1) and 15 (M2). The hard-coded list should be regenerated from these. Note that site 47 is selected by the heuristics but not by the base or +Size MIQP on either matrix, so it sits in the within-matrix backbones but not the cross-matrix one.
- `scripts/prepare_miqp_data.py` contains a commented-out `np.fill_diagonal(P, 0.0)` with a “remove self-loops” comment. Self-recruitment is intentionally retained, so that line should stay disabled and the comment reworded to avoid confusion.
- `scripts/experimental/run_miqp_inline.py` and `run_miqp_test.py` *do* zero the diagonal. They are not part of the canonical pipeline; either delete the `experimental/` folder or comment those lines so the diagonal-zeroed surrogate is never run by accident.
- `ampl/oyster_comm.dat` uses integer indices 0–48 (the position in `CANDIDATE_SITES`, not the original site label). Worth a one-line comment in the file since the same `.dat` is read alongside `.mod` files where $N = \{0, \dots, 48\}$.
- `runs/` contains ~ 60 timestamped greedy CSVs from the development iterations. A `runs/final/` subfolder with just the canonical CSVs would make the repo friendlier on first read.
- A LibreOffice / Office lock file `~$presentation.pptx` is checked in. Safe to delete.

8. Conclusions

Across two independent connectivity matrices, four MIQP variants, three ODE-driven heuristics, and two external forcing regimes — 18 distinct optimization runs in total — the same seven reefs are selected by every single one: 10, 31, 37, 40, 41, 49, 53. The eleven-site cross-matrix constant-forcing backbone $\{10, 15, 31, 32, 37, 40, 41, 49, 51, 52, 53\}$ is the next natural layer out, and the per-matrix backbones (17 sites for Matrix 1, 15 for Matrix 2) extend that further.

That this much agreement emerges from objectives that are conceptually quite different — the full nonlinear JARS equilibrium on one hand, a quadratic plumbing surrogate on the other — is not a tautology. It tells us that the surrogate captures the dominant structural drivers of the

metapopulation dynamics. From a planning perspective, the practical translation is: a small number of reefs do most of the connectivity work, and any restoration design that omits the seven-site backbone is, with high confidence, structurally worse than one that includes it.

A. All 18 Final Designs

For ease of cross-checking, every 25-site set used in this report is reproduced below as a flat list.

Model	Selected sites (sorted)
M1 Greedy (constant)	4, 10, 15, 16, 17, 19, 20, 21, 26, 27, 28, 30, 31, 32, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54
M1 Greedy + Swap (constant)	10, 12, 15, 16, 17, 20, 21, 26, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54, 59
M1 Stingy (constant)	10, 12, 15, 16, 17, 20, 21, 26, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54, 59
M1 MIQP (constant)	10, 11, 12, 15, 16, 17, 20, 26, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 49, 51, 52, 53, 54, 59
M1 MIQP + Comm	10, 12, 15, 16, 17, 21, 26, 28, 29, 31, 32, 33, 37, 40, 41, 42, 44, 47, 48, 49, 51, 52, 53, 54, 59
M1 MIQP + Size	10, 11, 12, 15, 16, 17, 20, 26, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 49, 51, 52, 53, 54, 59
M1 MIQP + Comm + Size	10, 12, 15, 16, 17, 21, 26, 28, 29, 31, 32, 33, 37, 40, 41, 42, 44, 47, 49, 51, 52, 53, 54, 56, 59
M1 Greedy + Swap (realistic)	4, 6, 10, 15, 19, 20, 21, 24, 27, 30, 31, 32, 36, 37, 38, 39, 40, 41, 47, 49, 51, 52, 53, 55, 60
M1 Stingy (realistic)	4, 10, 15, 16, 17, 19, 20, 21, 26, 27, 31, 32, 33, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54, 59
M2 Greedy (constant)	4, 10, 11, 12, 15, 17, 19, 20, 21, 26, 27, 28, 29, 30, 31, 32, 36, 37, 40, 41, 47, 49, 51, 52, 53
M2 Greedy + Swap (constant)	10, 11, 12, 15, 16, 17, 19, 20, 26, 29, 31, 32, 33, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54, 59
M2 Stingy (constant)	10, 11, 12, 15, 16, 17, 19, 20, 26, 29, 31, 32, 33, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53, 54, 59
M2 MIQP (constant)	10, 11, 12, 15, 16, 17, 20, 26, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 49, 51, 52, 53, 54, 59
M2 MIQP + Comm	10, 11, 12, 15, 16, 17, 21, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 42, 47, 49, 51, 52, 53, 56, 59
M2 MIQP + Size	10, 11, 12, 15, 16, 17, 20, 26, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 44, 49, 51, 52, 53, 54, 59
M2 MIQP + Comm + Size	10, 11, 12, 15, 16, 21, 26, 27, 28, 29, 31, 32, 33, 36, 37, 40, 41, 42, 47, 48, 49, 51, 52, 53, 59

Model	Selected sites (sorted)
M2 Greedy + Swap (realistic)	1, 4, 6, 10, 18, 19, 20, 21, 24, 30, 31, 35, 36, 37, 38, 39, 40, 41, 42, 47, 49, 53, 55, 57, 60
M2 Stingy (realistic)	4, 6, 7, 10, 15, 16, 19, 20, 21, 24, 26, 27, 30, 31, 32, 36, 37, 40, 41, 44, 47, 49, 51, 52, 53

Contact

Marouf Paul — mmarouf@wm.edu / maroufpaul2@gmail.com (post-graduation)

Advisor: Prof. Rex Kincaid (William & Mary, Mathematics)

Biological collaborators: Prof. Leah Shaw (W&M), Prof. Rom Lipcius (VIMS)